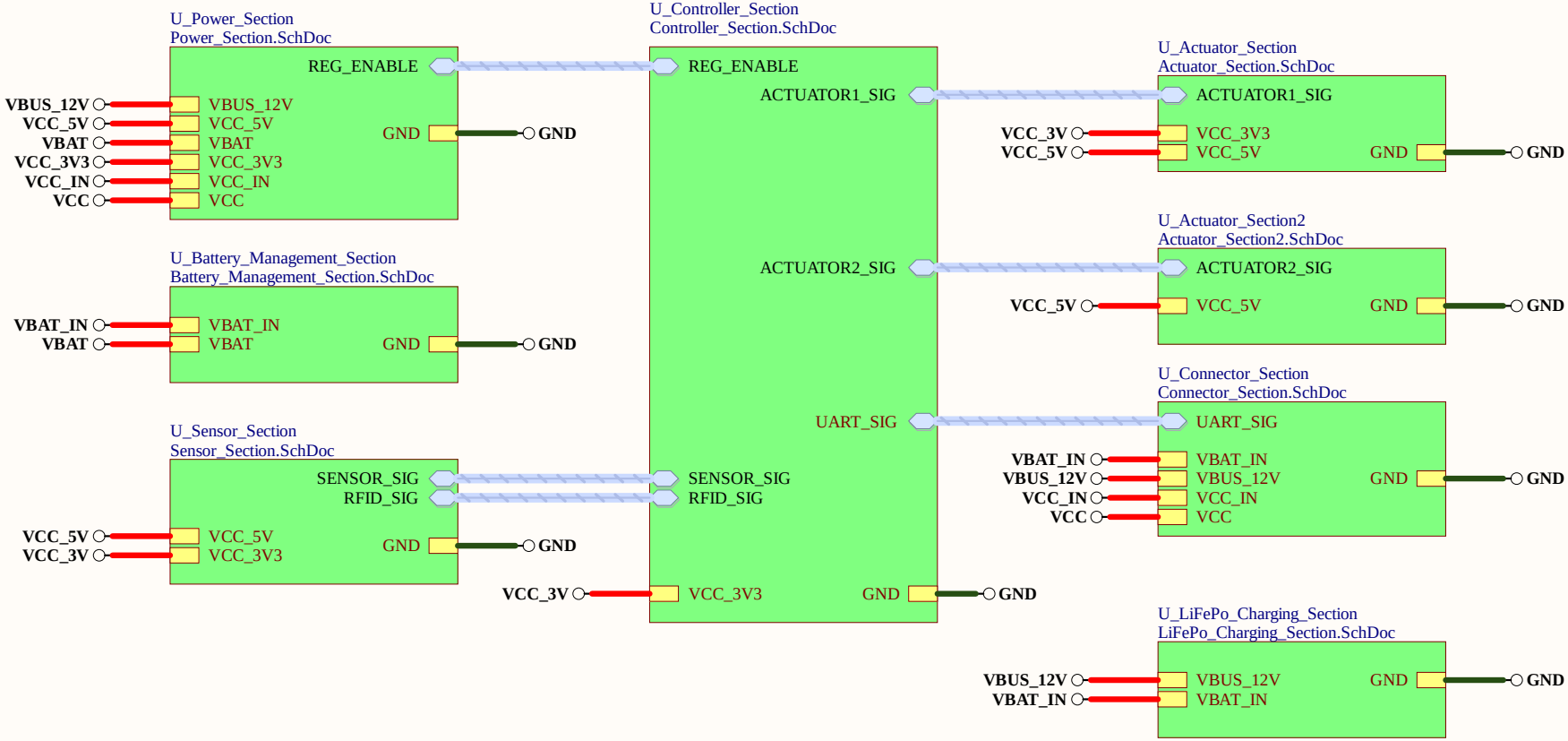
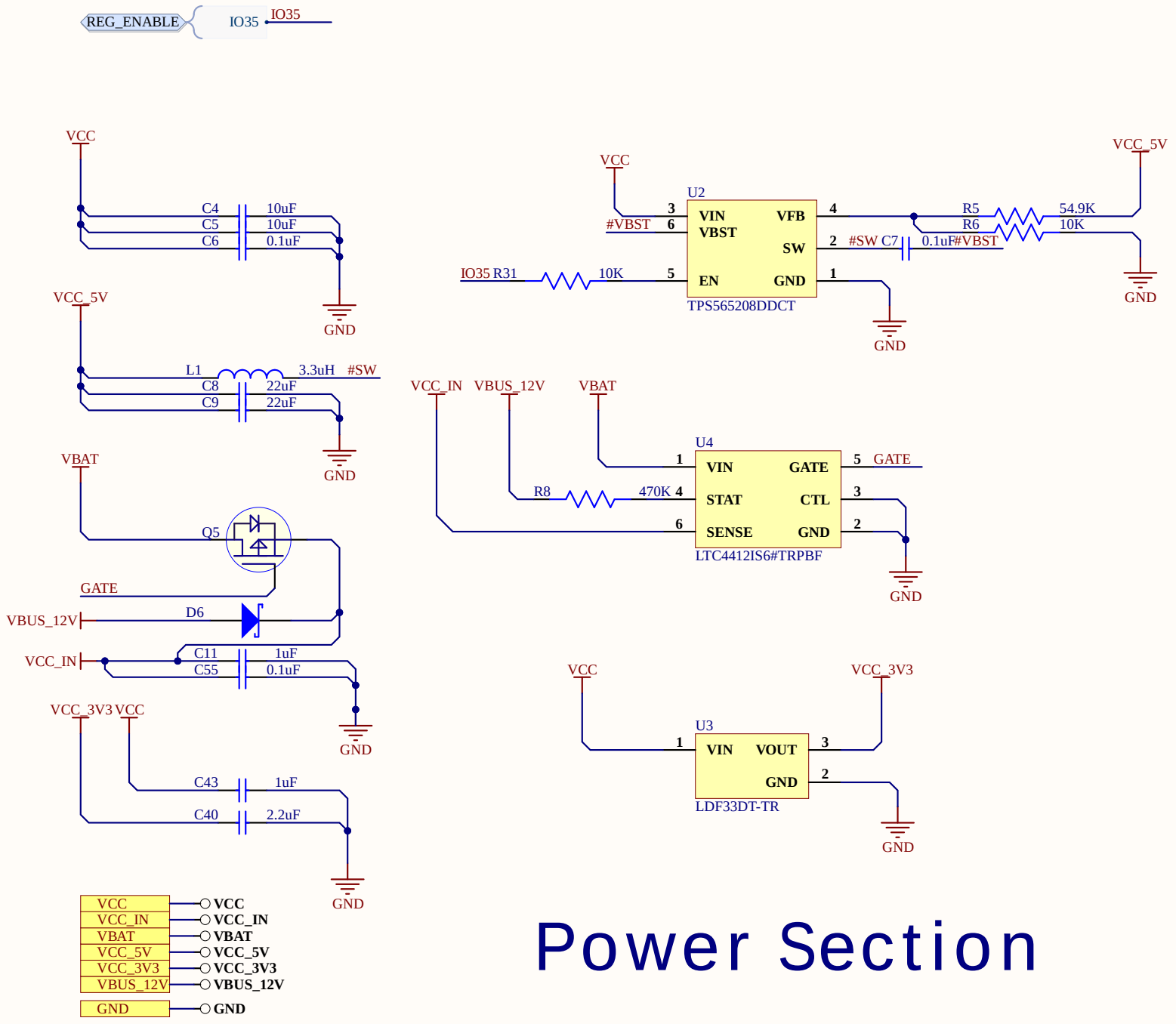
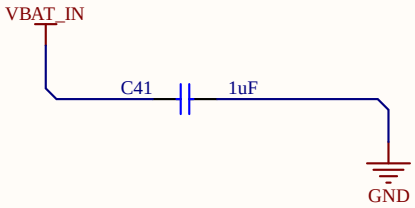
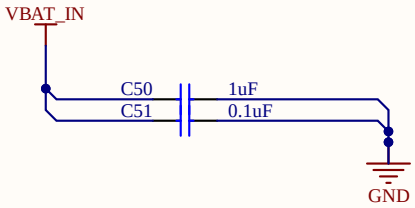
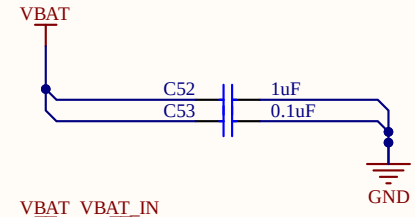
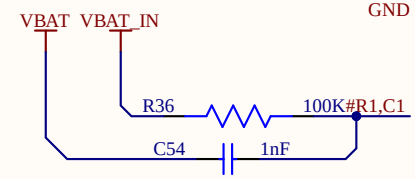

U_Power_Section
Power_Section.SchDoc
REG_ENABLE
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
GND
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
GND
U_Sensor_Section
Sensor_Section.SchDoc
SENSOR_SIG
RFID_SIG
VCC_5V
VCC_3V3
GND
U_Controller_Section
Controller_Section.SchDoc
REG_ENABLE
ACTUATOR1_SIG
ACTUATOR2_SIG
UART_SIG
SENSOR_SIG
RFID_SIG
VCC_3V3
GND
U_Actuator_Section
Actuator_Section.SchDoc
ACTUATOR1_SIG
VCC_3V3
VCC_5V
GND
U_Actuator_Section2
Actuator_Section2.SchDoc
ACTUATOR2_SIG
VCC_5V
GND
U_Connector_Section
Connector_Section.SchDoc
UART_SIG
VBAT_IN
VBUS_12V
VCC_IN
VCC
GND
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
GND

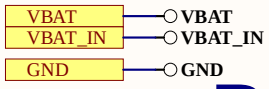

REG_ENABLE
IO35
IO35
VCC
C4 10uF
C5 10uF
C6 0.1uF
GND
VCC_5V
L1 3.3uH #SW
C8 22uF
C9 22uF
GND
VBAT
Q5
GATE
VBUS_12V
D6
VCC_IN
C11 1uF
C55 0.1uF
GND
VCC_3V3
VCC
C43 1uF
C40 2.2uF
GND
VCC
VCC_IN
VBAT
VCC_5V
VCC_3V3
VBUS_12V
GND
VCC
VCC_IN
VBAT
VCC_5V
VCC_3V3
VBUS_12V
GND
VCC
U2
3 VIN
6 VBST
4 VFB
2 SW
5 EN
1 GND
#VBST
IO35 R31 10K
TPS565208DDCT
#SW C7 0.1u
F#VBST
R5 54.9K
R6 10K
VCC_5V
GND
GND
VCC_IN
VBUS_12V
VBAT
U4
1 VIN
4 STAT
6 SENSE
5 GATE
3 CTL
2 GND
R8 470K
GATE
LTC4412IS6#TRPBF
GND
VCC
VCC_3V3
U3
1 VIN
3 VOUT
2 GND
LDF33DT-TR
GND
Power Section

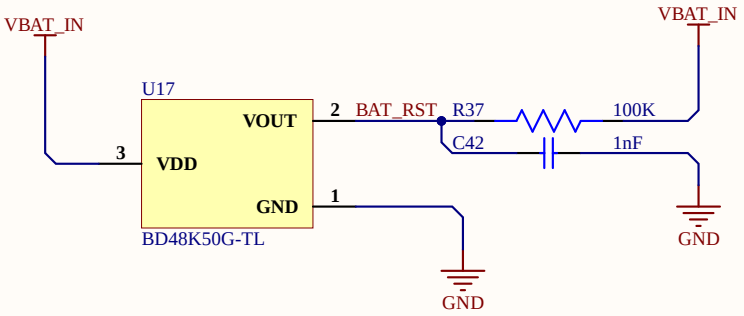


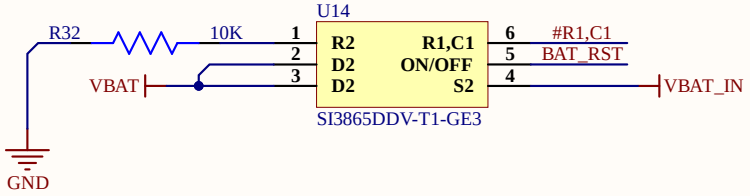




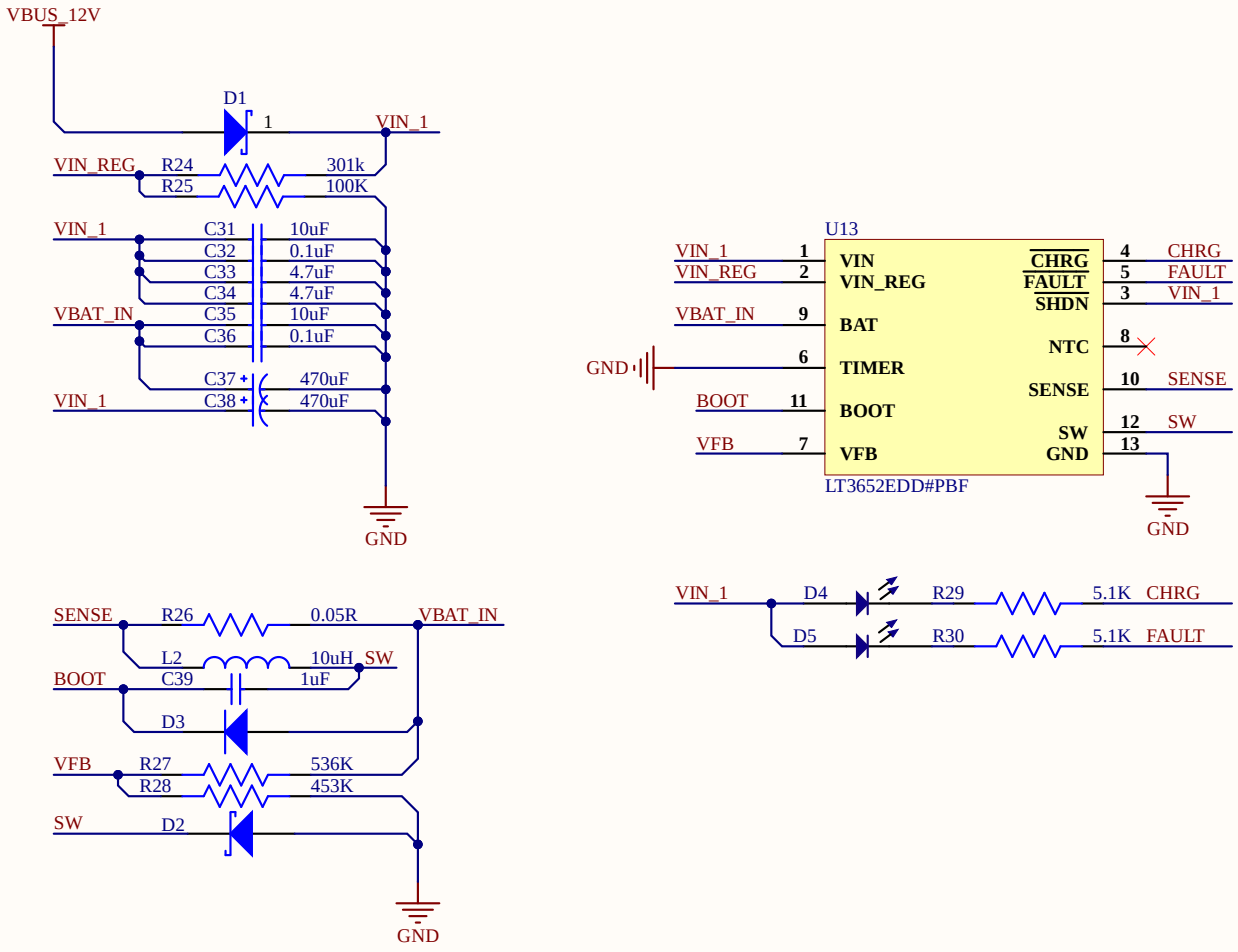




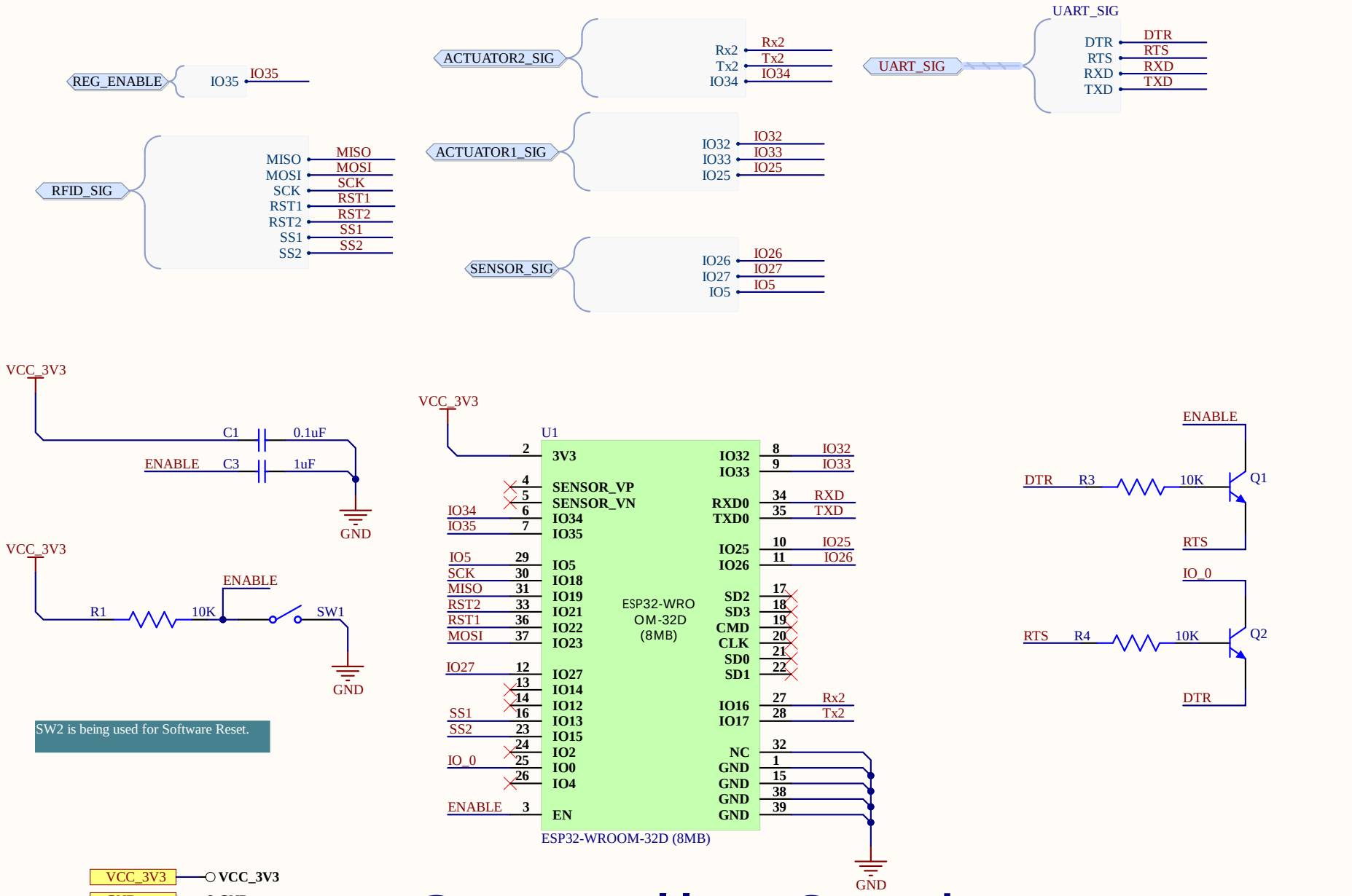




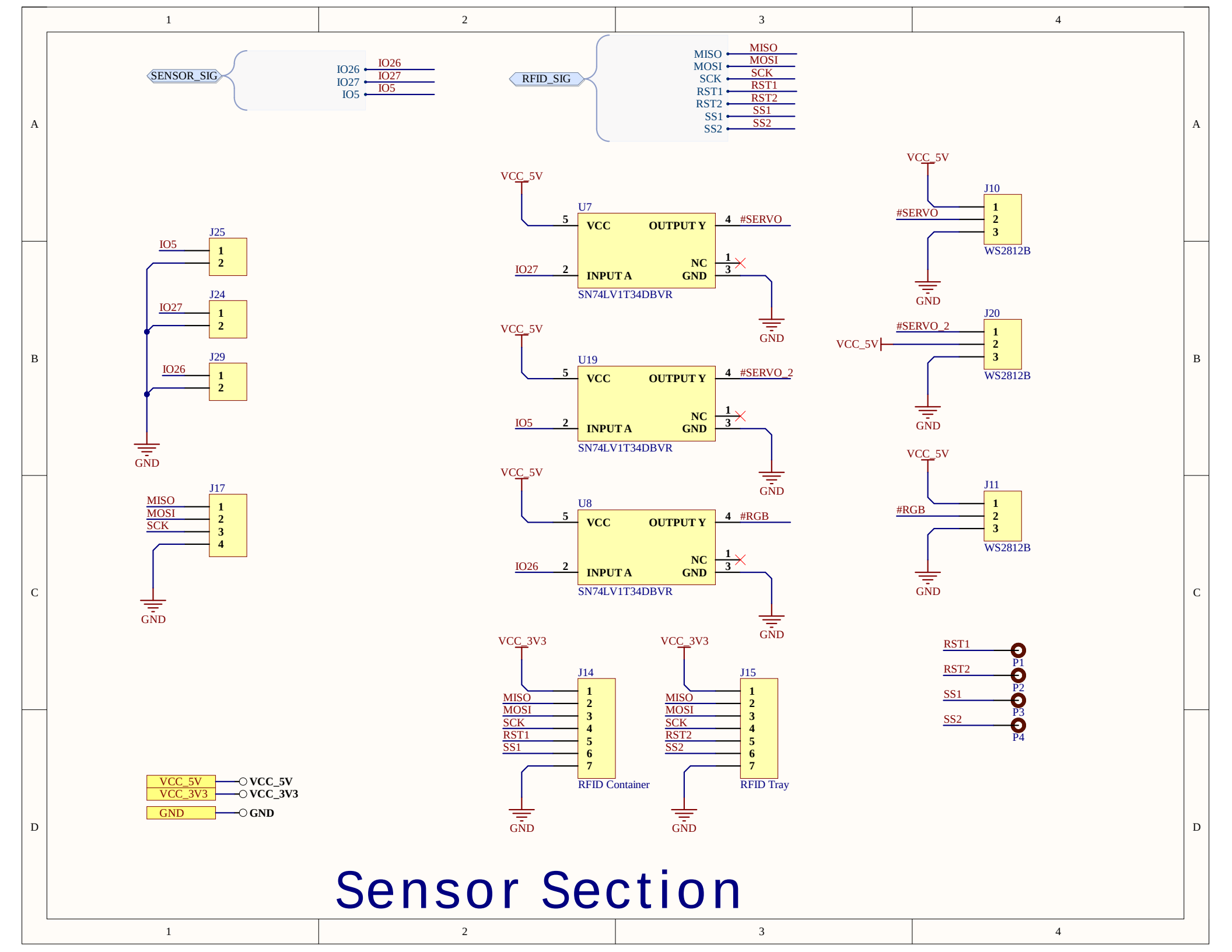
Battery Management Section



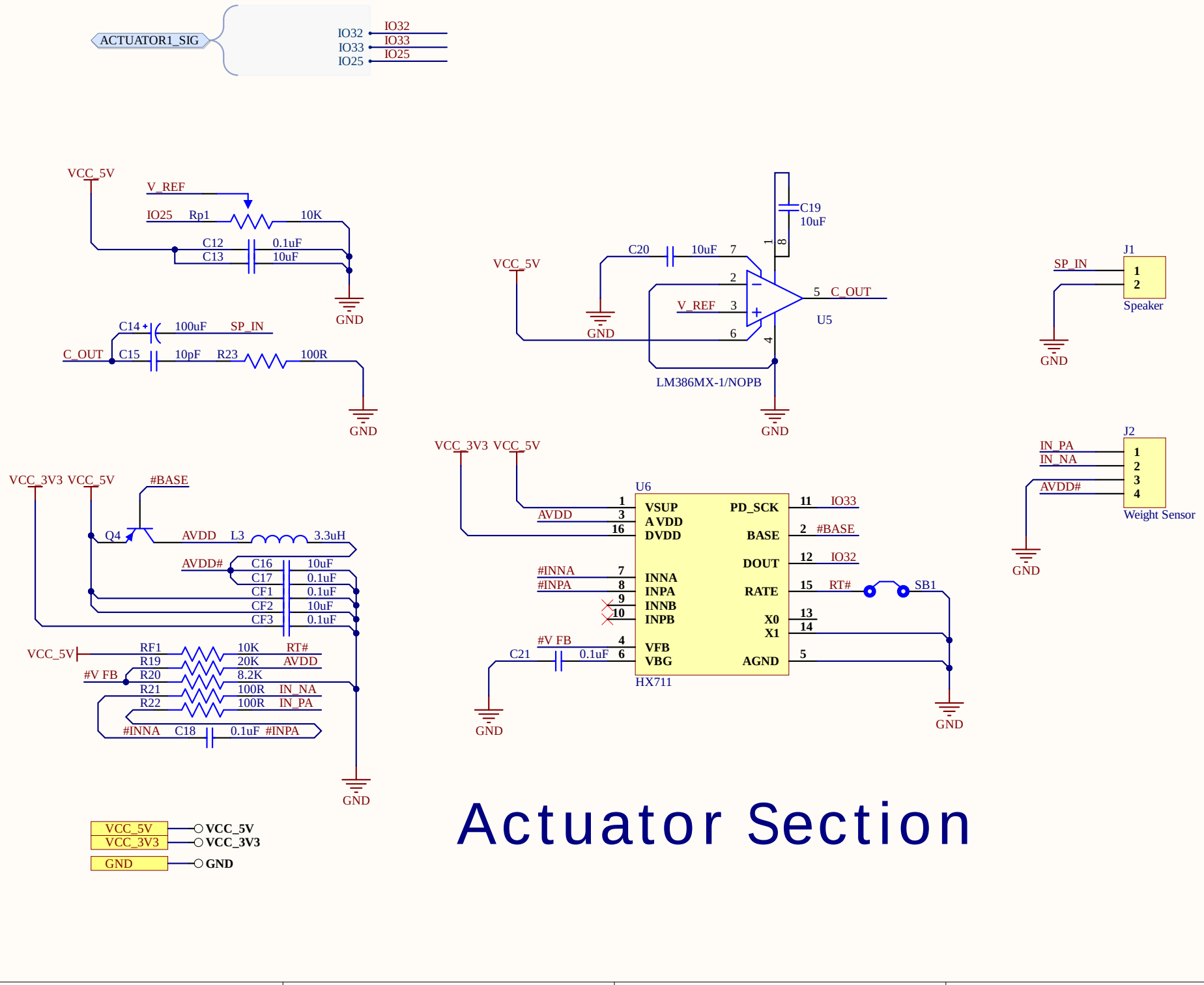
LiFePO4 Charging Section



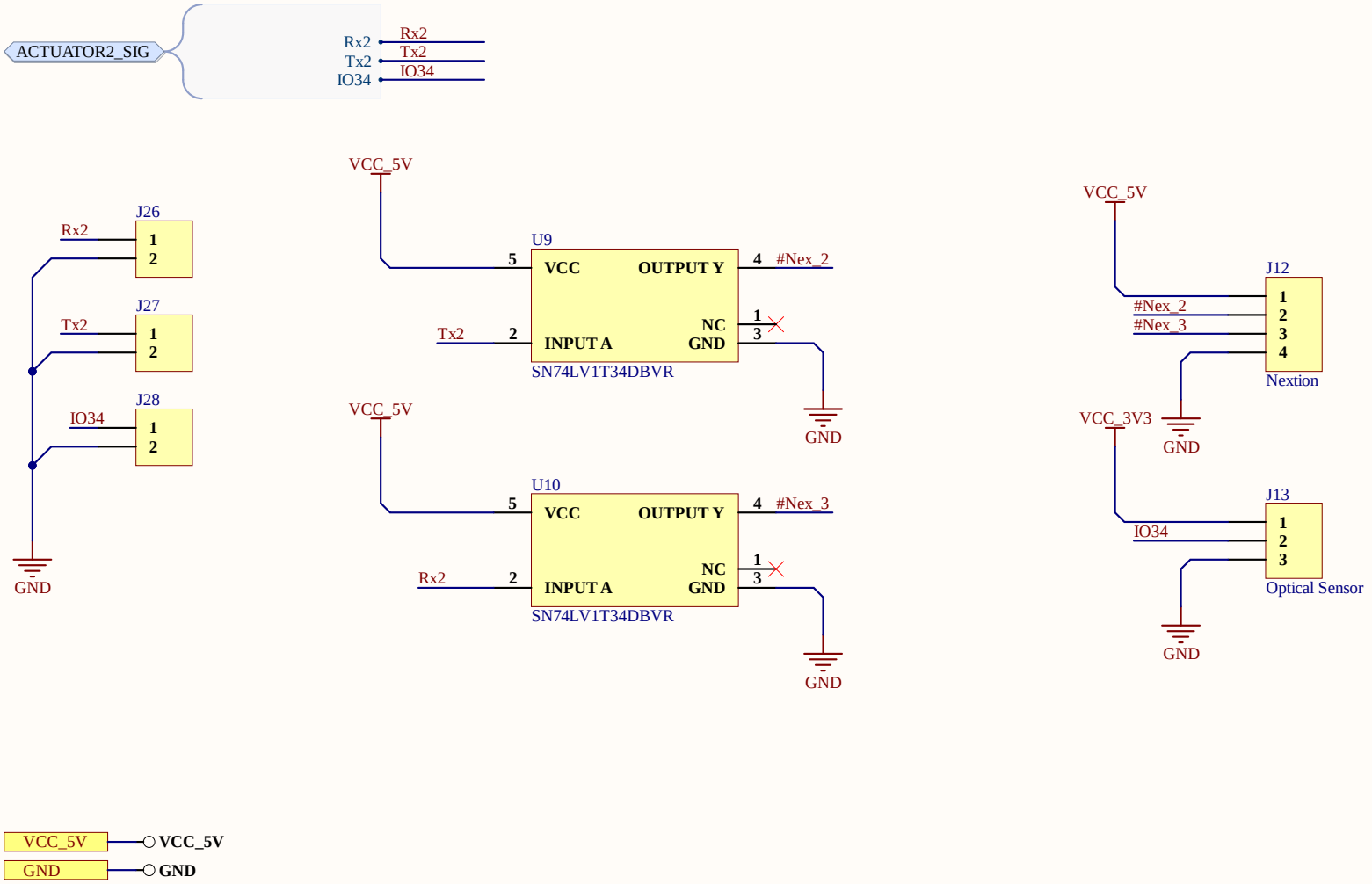
Controller Section



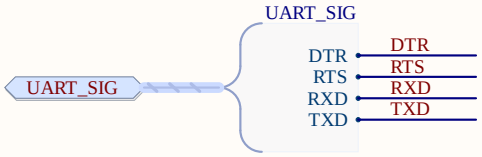
Sensor Section

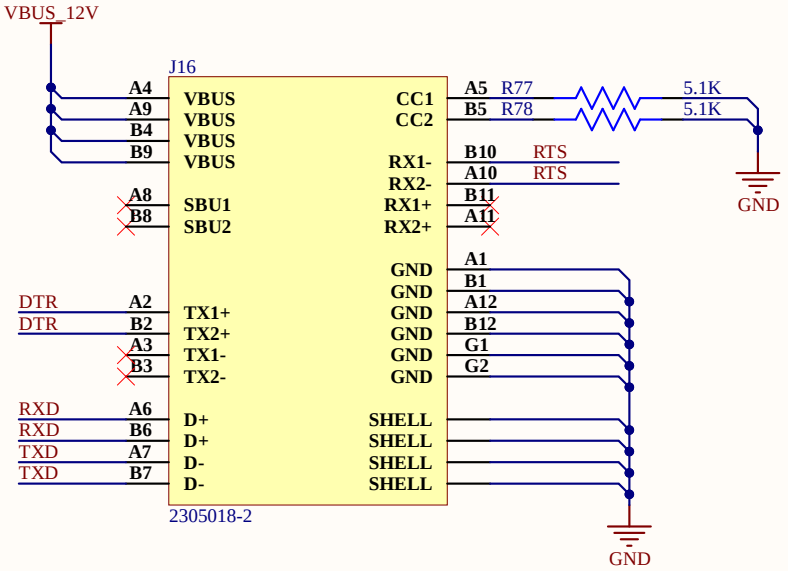


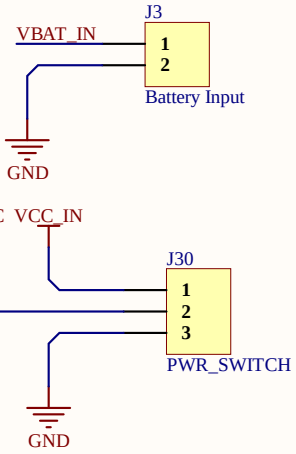
Actuator Section

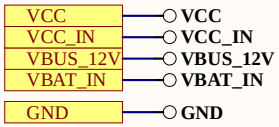


Actuator Section









Connector Section